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RADIAL ADJUSTING SELF-EXPANDING STENT WITH ANTI-MIGRATION FEATURES

Abstract

An esophageal stent may include an expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, the expandable framework being configured to expand from a radially collapsed configuration to a radially expanded configuration, and a polymeric outer sleeve disposed radially outward of and spaced apart from a body region of the expandable framework. The expandable framework may include a plurality of anti-migration supports extending radially outward from the body region of the expandable framework toward the polymeric outer sleeve.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/462,225, filed Sep. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/127,294, filed Dec. 18, 2020, now U.S. Pat. No. 11,786,355, which claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 62/967,755, filed on Jan. 30, 2020, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The disclosure is directed to a stent and a method for forming a stent. More particularly, the disclosure is directed to a self-expanding stent having anti-migration features and a method of forming a self-expanding stent having anti-migration features.

BACKGROUND

[0003] A stent may be configured to be positioned in a body lumen for a variety of medical applications. For example, a stent may be used to treat a stenosis in a blood vessel, used to maintain a fluid opening or pathway in the vascular, urinary, biliary, tracheobronchial, esophageal or renal tracts, or to position a device such as an artificial valve or filter within a body lumen, in some instances. In some cases, a stent may include anti-migration features in order to help anchor the stent in place in whichever body lumen the stent is placed. In some instances, forming these anti-migration features may be difficult to do accurately and repeatedly. Of the known medical devices and methods of manufacture, each has certain advantages and disadvantages. There is an ongoing need to provide alternative medical devices and methods of manufacture.

SUMMARY

[0004] In one example, an esophageal stent may comprise: an expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, the expandable framework being configured to expand from a radially collapsed configuration to a radially expanded configuration; and a polymeric outer sleeve disposed radially outward of and spaced apart from a body region of the expandable framework. The expandable framework may include a plurality of anti-migration supports extending radially outward from the body region of the expandable framework toward the polymeric outer sleeve.

[0005] In addition or alternatively to any example disclosed herein, the expandable framework is configured to self-expand from the radially collapsed configuration to the radially expanded configuration when unconstrained.

[0006] In addition or alternatively to any example disclosed herein, the plurality of anti-migration supports extends radially outward from the body region at an oblique angle to the central longitudinal axis.

[0007] In addition or alternatively to any example disclosed herein, the plurality of anti-migration supports is angled toward the second end of the expandable framework.

[0008] In addition or alternatively to any example disclosed herein, the oblique angle is greater than about 45 degrees and less than 90 degrees.

[0009] In addition or alternatively to any example disclosed herein, the expandable framework may

include a first flared region proximate the first end and a second flared region proximate the second end. The body region may extend from the first flared region to the second flared region.

[0010] In addition or alternatively to any example disclosed herein, the expandable framework is encapsulated within a polymeric covering.

[0011] In addition or alternatively to any example disclosed herein, the plurality of anti-migration supports is a plurality of anti-migration loops each hingedly attached to the expandable framework at a base of the anti-migration loops.

[0012] In addition or alternatively to any example disclosed herein, an esophageal stent may comprise: an expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, the expandable framework being configured to expand from a radially collapsed configuration to a radially expanded configuration; and a polymeric outer sleeve disposed radially outward of and spaced apart from a body region of the expandable framework. The expandable framework may include a plurality of interwoven filaments. At least a portion of the plurality of interwoven filaments may extend radially outward beyond an outer diameter of the body region to form anti-migration loops. The anti-migration loops may extend from the body region at a first angle. A base of each of the anti-migration loops may form a hinge portion such that the anti-migration loops are elastically deflectable from the first angle to a second angle less than the first angle. The anti-migration loops may extend radially outward from the body region to the polymeric outer sleeve.

[0013] In addition or alternatively to any example disclosed herein, the anti-migration loops are configured to deflect radially inward at the base when subjected to a radially inward force.

[0014] In addition or alternatively to any example disclosed herein, the expandable framework is configured to maintain a constant inner diameter when the anti-migration loops are subjected to the radially inward force.

[0015] In addition or alternatively to any example disclosed herein, an axial force applied to the expandable framework in a direction toward the second end in situ is transmitted radially outward through the anti-migration loops.

[0016] In addition or alternatively to any example disclosed herein, the expandable framework may include a first flared region proximate the first end and a second flared region proximate the second end. The body region may extend from the first flared region to the second flared region. The first flared region and the second flared region may have an outer diameter in the radially expanded configuration, and the outer diameter of the body region may be less than the outer diameter of the first flared region and the second flared region. An outer radial extent of the plurality of anti-migration loops in the radially expanded configuration may be substantially equal to the outer diameter of the first flared region and the second flared region.

[0017] In addition or alternatively to any example disclosed herein, the first angle is an oblique angle.

[0018] In addition or alternatively to any example disclosed herein, an esophageal stent may comprise: a metallic expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, the expandable framework being configured to self-expand from a radially collapsed configuration to a radially expanded configuration when unconstrained; and a polymeric outer sleeve disposed radially outward of the expandable framework. The expandable framework may include a first flared region proximate the first end, a second flared region proximate the second end, and a body region extending from the first flared region to the second flared region. The expandable framework may be encapsulated within a polymeric covering. The expandable framework may include a plurality of anti-migration loops each hingedly attached to the expandable framework at a base of the plurality of anti-migration loops. The plurality of anti-migration loops may be heat set to be biased toward the expanded configuration.

[0019] In addition or alternatively to any example disclosed herein, the polymeric covering and the

polymeric outer sleeve are fixedly attached together at the first flared region and the second flared region.

[0020] In addition or alternatively to any example disclosed herein, the plurality of anti-migration loops spaces the polymeric outer sleeve apart from the body region.

[0021] In addition or alternatively to any example disclosed herein, the body region maintains a fixed inner diameter when subjected to a radially inward force up to about 5 Newtons.

[0022] In addition or alternatively to any example disclosed herein, wherein the plurality of anti-migration loops is configured to deflect radially inward toward the body region at the base when subjected to a radially inward force less than about 5 Newtons.

[0023] In addition or alternatively to any example disclosed herein, each of the plurality of anti-migration loops is formed of an arcuate segment of a filament of the expandable framework, wherein the arcuate segment of the filament forming the anti-migration loop is located between first and second segments of the filament helically extending around the expandable framework interwoven with other filaments of the expandable framework.

[0024] The above summary of some embodiments, aspects, and/or examples is not intended to describe each embodiment or every implementation of the present disclosure. The figures and the detailed description which follows more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The invention may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:

[0026] FIG. 1 is a schematic partial cross-sectional view of an esophageal stent disposed within a delivery sheath in a radially collapsed configuration;

[0027] FIG. 2 illustrates selected aspects of an expandable framework of an esophageal stent in a radially expanded configuration;

[0028] FIGS. 3-4 illustrate some examples of the formation of the plurality of anti-migration supports or loops;

[0029] FIG. 5 illustrates selected aspects of an esophageal stent;

[0030] FIG. 6 is a schematic partial cross-sectional view of an esophageal stent disposed within a body lumen in a radially expanded configuration; and

[0031] FIGS. 7-8 illustrate selected aspects of the esophageal stent being subjected to peristaltic forces within the body lumen.

[0032] While the invention is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention.

DETAILED DESCRIPTION

[0033] The following description should be read with reference to the drawings, which are not necessarily to scale, wherein like reference numerals indicate like elements throughout the several views. The detailed description and drawings are intended to illustrate but not limit the claimed invention. Those skilled in the art will recognize that the various elements described and/or shown may be arranged in various combinations and configurations without departing from the scope of the disclosure. The detailed description and drawings illustrate example embodiments of the claimed invention. However, in the interest of clarity and ease of understanding, while every feature and/or element may not be shown in each drawing, the feature(s) and/or element(s) may be understood to be present regardless, unless otherwise specified.

[0034] For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

[0035] All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about”, in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (e.g., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.

[0036] The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

[0037] Although some suitable dimensions, ranges, and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges, and/or values may deviate from those expressly disclosed.

[0038] As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise. It is to be noted that in order to facilitate understanding, certain features of the disclosure may be described in the singular, even though those features may be plural or recurring within the disclosed embodiment(s). Each instance of the features may include and/or be encompassed by the singular disclosure(s), unless expressly stated to the contrary. For simplicity and clarity purposes, not all elements of the disclosed invention are necessarily shown in each figure or discussed in detail below. However, it will be understood that the following discussion may apply equally to any and/or all of the components for which there are more than one, unless explicitly stated to the contrary.

[0039] Additionally, not all instances of some elements or features may be shown in each figure for clarity.

[0040] Relative terms such as “proximal”, “distal”, “advance”, “retract”, variants thereof, and the like, may be generally considered with respect to the positioning, direction, and/or operation of various elements relative to a user/operator/manipulator of the device, wherein “proximal” and “retract” indicate or refer to closer to or toward the user and “distal” and “advance” indicate or refer to farther from or away from the user. In some instances, the terms “proximal” and “distal” may be arbitrarily assigned in an effort to facilitate understanding of the disclosure, and such instances will be readily apparent to the skilled artisan. Other relative terms, such as “upstream”, “downstream”, “inflow”, and “outflow” refer to a direction of fluid flow within a lumen, such as a body lumen, a blood vessel, or within a device.

[0041] It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect the particular feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

[0042] For the purpose of clarity, certain identifying numerical nomenclature (e.g., first, second,

third, fourth, etc.) may be used throughout the description and/or claims to name and/or differentiate between various described and/or claimed features. It is to be understood that the numerical nomenclature is not intended to be limiting and is exemplary only. In some embodiments, alterations of and deviations from previously used numerical nomenclature may be made in the interest of brevity and clarity. That is, a feature identified as a “first” element may later be referred to as a “second” element, a “third” element, etc. or may be omitted entirely, and/or a different feature may be referred to as the “first” element. The meaning and/or designation in each instance will be apparent to the skilled practitioner.

[0043] FIG. 1 schematically illustrates an esophageal stent **10** disposed within a delivery sheath **90**. To improve clarity, some features of the esophageal stent **10** disclosed herein are not shown in FIG. 1, such as a polymeric covering **22** and/or a polymeric outer sleeve **30** (e.g., FIGS. 5-8). The delivery sheath **90** may radially constrain the esophageal stent **10** in a radially collapsed configuration. As shown in FIG. 1, the esophageal stent **10** may be disposed within a distal portion of a lumen of the delivery sheath **90**. The delivery sheath **90** is illustrated with a distal opening in communication with the lumen. Other configurations are also contemplated. In some embodiments, delivery of the esophageal stent **10** may involve retracting the delivery sheath **90** relative to the esophageal stent **10** or advancement of the esophageal stent **10** out of the delivery sheath **90**. In some embodiments, an internal member (not shown) may be disposed within the lumen of the delivery sheath **90**. For example, the internal member may be configured to hold the esophageal stent **10** as the delivery sheath **90** is retracted or the internal member may be configured to push or advance the esophageal stent **10** out of the distal opening of the delivery sheath **90**. Other configurations are also contemplated.

[0044] As seen in FIG. 2, the esophageal stent **10** may include an expandable framework **11** having a first end **12**, a second end **14** and a lumen extending therethrough. In at least some embodiments, the expandable framework **11** may be and or define a generally tubular structure. The esophageal stent **10** and/or the expandable framework **11** may include a central longitudinal axis extending from the first end **12** to the second end **14**. The expandable framework **11** may be configured to expand and/or self-expand from the radially collapsed configuration (e.g., FIG. 1) to a radially expanded configuration (e.g., FIG. 2). In some embodiments, the expandable framework **11** may be configured to self-expand from the radially collapsed configuration to the radially expanded configuration when unconstrained (e.g., radially and/or axially unconstrained). In some embodiments, the expandable framework **11** in the radially collapsed configuration may be elongated and/or longer than in the radially expanded configuration. As such, in some embodiments, when the expandable framework **11** is radially expanded, the expandable framework **11** may shorten axially.

[0045] In at least some embodiments, the expandable framework **11** may include a first flared region **13** proximate the first end **12** and a second flared region **15** proximate the second end **14**. The expandable framework **11** may include a body region **17**. In some embodiments, the body region **17** may extend from the first flared region **13** to the second flared region **15**. In some embodiments, the first flared region **13** and/or the second flared region **15** have an outer diameter in the radially expanded configuration, wherein an outer diameter of the body region **17** is less than the outer diameter of the first flared region **13** and/or the second flared region **15**.

[0046] In some embodiments, the expandable framework **11** may include a plurality of interwoven filaments. In some embodiments, the plurality of interwoven filaments may include a plurality of first filaments **16** each extending in a first helical path around the central longitudinal axis in a first direction from the first end **12** toward and/or to the second end **14**. In some embodiments, the first direction may be clockwise. In some embodiments, the plurality of interwoven filaments may include a plurality of second filaments **18** each extending in a second helical path around the central longitudinal axis in a second direction from the first end **12** toward and/or to the second end **14**. In some embodiments, the second direction may be opposite the first direction. In some

embodiments, the second direction may be counterclockwise. In some embodiments, the plurality of first filaments **16** may be interwoven with the plurality of second filaments **18**, such as when forming a braided stent (e.g., FIG. 3). Alternatively, in some embodiments, the expandable framework **11** may be a monolithic structure including a plurality of interconnected struts and/or a single filament interwoven with itself, such as when forming a knitted stent (e.g., FIG. 4). In some embodiments, the expandable framework **11** may be cut from a tubular member, using various means known in the art (e.g., laser, machining, etching, chemical dissolution, etc.). In at least some embodiments, the expandable framework **11** may be formed from a metallic material, a polymeric material, a composite material, combinations thereof, etc. Some examples of suitable but non-limiting materials for the expandable framework **11** and/or elements or components thereof are described below.

[0047] The expandable framework **11** may include a plurality of anti-migration supports **20** (e.g., arms, struts, projections, etc.) extending radially outward from the body region **17** of the expandable framework **11**. In some embodiments, the plurality of anti-migration supports **20** may be a plurality of anti-migration loops each hingedly and/or pivotably attached to the expandable framework **11** at a base **21** of the plurality of anti-migration loops and/or the plurality of anti-migration supports **20**. The plurality of anti-migration supports **20** and/or the anti-migration loops may extend from the base **21** toward the second end **14** of the expandable framework **11**. In some embodiments, at least a portion of the plurality of interwoven filaments may extend radially outward beyond the outer diameter of the body region **17** of the esophageal stent **10** to form the plurality of anti-migration supports **20** and/or the anti-migration loops, as seen in the examples of FIGS. 3 and 4. Other configurations and/or examples are also contemplated. In at least some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be formed with the expandable framework **11** as a single monolithic structure. For instance, the expandable framework **11** may be formed of a plurality of interwoven (e.g., braided) filaments. In some instances, the interwoven filaments forming the expandable framework **11** may extend helically around the expandable framework **11** and cross over and/or under filaments extending helically in an opposite direction. Each of the plurality of anti-migration loops may be formed of an integral segment, such as an arcuate segment, of a filament of the expandable framework **11**, wherein the arcuate segment of the filament forming the anti-migration loop is located between first and second segments of the filament helically extending around the expandable framework **11** interwoven with other filaments of the expandable framework **11**.

[0048] The plurality of anti-migration supports **20** and/or the anti-migration loops may be spaced apart from one another longitudinally along the length of the body region **17** of the expandable framework **11** and/or circumferentially around the circumference of the body region **17** of the expandable framework **11**. For instance, the body region **17** of the expandable framework **11** may include 2 or more, 3 or more, 4 or more, 5 or more, or 6 or more longitudinal rows, helical rows and/or circumferential rows of anti-migration supports **20** and/or anti-migration loops, with 5 or more, 8 or more, 10 or more, 14 or more, or 20 or more anti-migration supports **20** and/or anti-migration loops in each row. In the figures, the plurality of anti-migration supports **20** in a given row are illustrated as being aligned with the plurality of anti-migration supports **20** in adjacent rows, but this is not strictly necessary in all embodiments. For example, the plurality of anti-migration supports **20** may be staggered or offset from each other in any given pair of adjacent rows.

[0049] In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may extend radially outward of and/or from the body region **17** of the expandable framework **11** about 2.0 millimeters (mm), about 2.5 mm, about 3.0 mm, about 3.5 mm, about 4.0 mm, about 4.5 mm, about 5.0 mm, about 5.5 mm, about 6.0 mm, about 6.5 mm, about 7.0 mm, about 7.5 mm, about 8.0 mm, or another suitable distance, in the expanded configuration. In some exemplary embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may

extend radially outward from the body region **17** about 4.0 mm to about 6.0 mm in the expanded configuration.

[0050] In some embodiments, the esophageal stent **10** may be a covered stent. In some embodiments, the esophageal stent **10** may include a polymeric outer sleeve **30** disposed radially outward of the expandable framework **11**. In some embodiments, the polymeric outer sleeve **30** may be spaced apart from at least the body region **17** of the expandable framework **11**, as seen in FIG. 5. It should be noted that FIGS. 5-8 are merely illustrative, and the expandable framework **11** may include more or fewer anti-migration supports and/or rows of anti-migration supports than shown. In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may extend radially outward from the body region **17** of the expandable framework **11** toward and/or to the polymeric outer sleeve **30**. In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops spaces the polymeric outer sleeve **30** apart from at least the body region **17** of the expandable framework **11**, as seen in FIG. 6, which illustrates the esophageal stent **10** disposed within a body lumen **100** (e.g., the esophagus) in partial cross-section. In some embodiments, the polymeric outer sleeve **30** may be attached to the expandable framework **11** at the first flared region **13** and/or the second flared region **15**.

[0051] In some embodiments, the expandable framework **11** may be encapsulated within a polymeric covering **22**. The polymeric covering **22** may span interstices in the expandable framework **11** and/or may define a lumen extending through the esophageal stent **10** and/or the expandable framework **11**. In some embodiments, the polymeric covering **22** may encapsulate, be disposed on and/or attached to the plurality of first filaments and the plurality of second filaments. In some embodiments, the polymeric covering **22** may span interstices between adjacent filaments of the plurality of first filaments and the plurality of second filaments. In some instances, the polymeric covering **22** may fully cover the expandable framework **11** along the entire length and circumference of the expandable framework **11**.

[0052] In some embodiments, the polymeric covering **22** and/or the polymeric outer sleeve **30** may be impervious to fluids, debris, and/or tissue ingrowth. In some embodiments, the polymeric covering **22** and/or the polymeric outer sleeve **30** may extend along the body region **17** of the expandable framework **11** from the first flared region **13** to the second flared region **15**. In some embodiments, the polymeric covering **22** and/or the polymeric outer sleeve **30** may extend along an entire length and circumference of the esophageal stent **10** and/or the expandable framework **11** from the first end **12** to the second end **14**. In some embodiments, the polymeric covering **22** may be disposed on an inner surface of the body region **17**, an outer surface of the body region **17**, both the inner surface and the outer surface of the body region **17**, or the body region **17** may be embedded within the polymeric covering **22** with the plurality of anti-migration supports **20** and/or the anti-migration loops protruding radially outward from the polymeric covering **22**. Other configurations are also contemplated.

[0053] In some embodiments, the body region **17**, the first flared region **13**, and/or the second flared region **15** may be encapsulated within a first portion of the polymeric covering **22**. In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be encapsulated within a second portion **24** of the polymeric covering **22**. In some embodiments, the entire expandable framework **11** may be encapsulated within the polymeric covering **22**. In some embodiments, the first portion of the polymeric covering **22** and the second portion **24** of the polymeric covering **22** may be formed as a single monolithic structure from a single material. In some embodiments, the first portion of the polymeric covering **22** may be fixedly attached to the second portion **24** of the polymeric covering **22** at and/or adjacent the base **21** of the plurality of anti-migration supports **20** and/or the anti-migration loops. In some embodiments, the polymeric covering **22** and the polymeric outer sleeve **30** may be fixedly attached together at the first flared region **13** and/or the second flared region **15**. In some embodiments, the second portion **24** of the polymeric covering **22** may be fixedly attached to the polymeric outer sleeve **30** at a radially

outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops and/or the second portion **24** of the polymeric covering **22** disposed thereon. In some embodiments, the polymeric outer sleeve **30** may be a third portion of the polymeric covering **22**. In some embodiments, the first portion of the polymeric covering **22**, the second portion **24** of the polymeric covering **22**, and the third portion of the polymeric covering **22** (e.g., the polymeric outer sleeve **30**) may be formed as a single monolithic structure from a single material.

[0054] In some embodiments, the polymeric covering **22** (and/or the polymeric outer sleeve **30**) may be formed having a single Shore hardness. In some embodiments, each portion of the polymeric covering **22** may have a different Shore hardness. In some embodiments, the first portion of the polymeric covering **22** may have a different Shore hardness than the second portion **24** of the polymeric covering **22** and/or the third portion of the polymeric covering **22** (e.g., the polymeric outer sleeve **30**). In some embodiments, the Shore hardness of the second portion **24** of the polymeric covering **22** may be higher than the Shore hardness of the first portion of the polymeric covering **22**. Various combinations of Shore hardness (e.g., same, similar, and/or different) among these portions of the polymeric covering **22** are also contemplated.

[0055] In some embodiments, a plurality of spaces may be disposed between the polymeric covering **22** and the polymeric outer sleeve **30**. In some embodiments, a plurality of spaces may be disposed between adjacent second portions **24** of the polymeric covering **22** (e.g., between adjacent anti-migration supports **20** and/or between adjacent anti-migration loops). In some embodiments, the plurality of spaces may be connected to each other between the polymeric covering **22** and the polymeric outer sleeve **30** to define a collective space therebetween. In some embodiments, the plurality of spaces may be a plurality of discrete cells or chambers. Other configurations, including but not limited to combinations of these configurations, are also contemplated.

[0056] The plurality of anti-migration supports **20** and/or the anti-migration loops may extend radially outward from the body region **17** at a first angle $\theta_{\text{sub.1}}$ relative to the body region **17** and/or the central longitudinal axis when not subjected to an external force such as a peristaltic force, as seen in FIG. **6**. The plurality of anti-migration supports **20** and/or the anti-migration loops may be angled toward the second end **14** of the expandable framework **11**. In some embodiments, the first angle $\theta_{\text{sub.1}}$ may be an oblique angle. In some embodiments, the oblique angle may be greater than about 30 degrees and less than 90 degrees, greater than about 45 degrees and less than 90 degrees, greater than about 60 degrees and less than 90 degrees, or another angle suitable for and/or consistent with the characteristics described herein.

[0057] In some embodiments, a base **21** of each of the plurality of anti-migration supports **20** and/or the anti-migration loops may form a hinge portion such that the plurality of anti-migration supports **20** and/or the anti-migration loops may be elastically deflectable at the base **21** and/or the hinge portion from the first angle $\theta_{\text{sub.1}}$ to a second angle $\theta_{\text{sub.2}}$ less than the first angle $\theta_{\text{sub.1}}$ when subjected to an external force, such as a radially inwardly directed peristaltic force (e.g., FIGS. **7** and **8**). The second angle $\theta_{\text{sub.2}}$ may be 10 or more degrees less than the first angle $\theta_{\text{sub.1}}$, 15 or more degrees less than the first angle $\theta_{\text{sub.1}}$, 20 or more degrees less than the first angle $\theta_{\text{sub.1}}$, 30 or more degrees less than the first angle $\theta_{\text{sub.1}}$, or 45 or more degrees less than the first angle $\theta_{\text{sub.1}}$, for example.

[0058] In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be heat set (e.g., during manufacturing) to be self-biased toward the radially expanded configuration in which the anti-migration supports **20** and/or the anti-migration loops assume the first angle $\theta_{\text{sub.1}}$. Similarly, the esophageal stent **10** and/or the expandable framework **11** may be heat set to be self-biased toward the radially expanded configuration. In the radially expanded configuration, the expandable framework **11** and/or the body region **17** may define an inner diameter D defining the lumen extending therethrough.

[0059] In use, the esophageal stent **10** may be disposed within the body lumen **100** in the radially expanded configuration, as shown in FIG. **6**. In at least some embodiments, placement of the

esophageal stent **10** may be via conventional means known in the art. The polymeric outer sleeve **30** may be disposed against and/or in contact with an inner surface of the body lumen **100**. The polymeric outer sleeve **30** may prevent tissue ingrowth into the esophageal stent **10**, thereby facilitating easier removal of the esophageal stent **10** from the body lumen **100** at a later time. The first flared region **13** and/or the second flared region **15** may be self-biased toward the radially expanded configuration, and in the radially expanded configuration, may exert a radially outward force against the inner surface of the body lumen **100**.

[0060] Since the expandable framework **11** may be encapsulated within the polymeric covering **22**, a bolus may be able to easily pass through the lumen of the esophageal stent **10**. In an instance where the bolus does not pass easily through the lumen of the esophageal stent **10**, axial forces may be exerted upon and/or applied to the esophageal stent **10** and/or the expandable framework **11** in a downstream direction toward the second end. Since the plurality of anti-migration supports **20** and/or the anti-migration loops are hingedly and/or pivotably attached to the expandable framework **11** at the base **21** of the plurality of anti-migration supports **20** and/or the anti-migration loops, axial forces applied to the esophageal stent **10** and/or the expandable framework **11** may cause the plurality of anti-migration supports **20** and/or the anti-migration loops, which are substantially rigid in a direction parallel to the central longitudinal axis between the base **21** and the radially outermost extent, to hinge or pivot radially outward at the base **21**, thereby exerting and/or increasing a radially outward force exerted on the body lumen **100** due to the plurality of anti-migration supports **20** and/or the anti-migration loops being wedged between the inner surface of the body lumen **100** and the body region **17** of the expandable framework **11**, which may resist radially inward collapse due to stent construction, heat setting, self-bias, and/or other factors. As such, axial forces may be transmitted radially outward through the plurality of anti-migration supports **20** and/or the anti-migration loops to the inner surface of the body lumen **100**, thereby increasing an anchoring force and/or increasing resistance of the esophageal stent **10** to axial migration within the body lumen **100**.

[0061] Some body lumens may exert peristaltic forces or motion on and/or against a stent disposed therein. The peristaltic forces or motion may include an axial component. In some instances, repeated exposure to the peristaltic forces or motion may cause a stent to migrate axially downstream within the body lumen. Migration is typically undesirable as placement is usually intended to treat a specific area or location within the body lumen. FIGS. 7-8 illustrate peristaltic forces or motion passing by the esophageal stent **10** disposed within the body lumen **100**. As may be seen in the figures, the peristaltic forces or motion may apply a radially inward force F against the esophageal stent **10**. The plurality of anti-migration supports **20** and/or the anti-migration loops may be configured to elastically deflect radially inward at the base **21** from a first angle $\theta_{sub.1}$ to a second angle $\theta_{sub.2}$ less than the first angle $\theta_{sub.1}$ while the body region **17** of the expandable framework **11** maintains a constant inner diameter. As the radially inward force F passes and/or is relieved, the plurality of anti-migration supports **20** and/or the anti-migration loops may elastically return to the first angle $\theta_{sub.1}$, as seen in FIG. 8. The deflection characteristics of the plurality of anti-migration supports **20** and/or the anti-migration loops may reduce the axial component of the peristaltic forces or motion on the esophageal stent **10**, while maintaining anchoring and/or anti-migration forces on and/or against the inner surface of the body lumen **100**, thereby preventing axial migration of the esophageal stent **10** within the body lumen **100**.

[0062] In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be configured to deflect radially inward at the base **21** and/or the hinge portion when subjected to a radially inward force F , such as a radially inwardly directed peristaltic force, as seen in FIGS. 7-8. In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be configured to deflect radially inward toward the body region **17** at the base **21** and/or the hinge portion when subjected to a radially inward force F greater than about 0.5 Newtons and less than about 5.0 Newtons, greater than about 1.0 Newton and less than about 4

Newtons, greater than about 2.0 Newtons and less than about 3.0 Newtons, about 2.4 Newtons, or another suitable amount of force. For example, the plurality of anti-migration supports **20** and/or the anti-migration loops may be configured to deflect at the base **21** and/or the hinge portion such that a radially outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops is deflected, moved, and/or translated toward the second end **14** while moving closer to the central longitudinal axis of the expandable framework **11** when the plurality of anti-migration supports **20** and/or the anti-migration loops is subjected to a radially inward force F , thereby changing orientation of the plurality of anti-migration supports **20** and/or the anti-migration loops from the first angle $\theta_{sub.1}$ to the second angle $\theta_{sub.2}$.

[0063] In some embodiments, the expandable framework **11** and/or the body region **17** of the expandable framework **11** is configured to maintain a substantially constant and/or fixed inner diameter D when the plurality of anti-migration supports **20** and/or the anti-migration loops are subjected to the radially inward force F . In some embodiments, when the plurality of anti-migration supports **20** and/or the anti-migration loops are subjected to the radially inward force F , the expandable framework **11** and/or the body region **17** of the expandable framework **11** may be configured to maintain the inner diameter D at at least 90% of its original inner diameter D (e.g., when the radially inward force F is absent), at at least 95% of its original inner diameter D (e.g., when the radially inward force F is absent), at at least 97% of its original inner diameter D (e.g., when the radially inward force F is absent), and/or at at least 99% of its original inner diameter D (e.g., when the radially inward force F is absent). Other configurations are also contemplated.

[0064] In some embodiments, the expandable framework **11** and/or the body region **17** of the expandable framework **11** is configured to maintain a substantially constant and/or fixed inner diameter D when the plurality of anti-migration supports **20** and/or the anti-migration loops are subjected to the radially inward force F up to about 5.0 Newtons. For example, a radially inward force F up to about 5.0 Newtons sufficient to deflect the plurality of anti-migration supports **20** and/or the anti-migration loops will have substantially no effect on the shape and/or form of the body region **17** and/or the expandable framework **11**. Other configurations are also contemplated. When the radially inward force F is removed, the plurality of anti-migration supports **20** and/or the anti-migration loops may elastically return to the first angle $\theta_{sub.1}$.

[0065] In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops may be substantially rigid from the base **21** and/or the hinge portion to the radially outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops in a direction parallel to the central longitudinal axis. For example, the plurality of anti-migration supports **20** and/or the anti-migration loops may be configured to deflect at the base **21** and/or the hinge portion but may be prevented from deflecting toward the first end **12** or the second end **14** at a location between the base **21** and/or the hinge portion and the radially outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops. As such, in some embodiments, an axial force applied to the expandable framework **11** in a direction toward the second end **14** in situ may be transmitted radially outward through the plurality of anti-migration supports **20** and/or the anti-migration loops. Since the plurality of anti-migration supports **20** and/or the anti-migration loops will not deflect or bend along their length (under expected in situ forces of less than 5.0 Newtons), the hinging and/or pivoting motion at the base **21** will cause additional radially outward force to be applied to a wall of a lumen, thereby increasing anchoring forces and reducing and/or preventing axial migration of the esophageal stent **10** in situ.

[0066] In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops extending radially outward from the body region **17** of the esophageal stent **10** and/or the expandable framework **11** may form a circumferential row of the plurality of anti-migration supports **20** and/or the anti-migration loops extending around the body region **17** of the expandable framework **11**. In some embodiments, the plurality of anti-migration supports **20** and/or the anti-migration loops within the circumferential row of the plurality of anti-migration supports **20** and/or

the anti-migration loops may be axially and/or circumferentially aligned at a common axial location along the central longitudinal axis of the esophageal stent **10** and/or the expandable framework **11**.

[0067] In some embodiments, an outer radial extent and/or the radially outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops in the radially expanded configuration and/or at the first angle $\theta_{sub.1}$ may be substantially equal to the outer diameter of the first flared region **13** and/or the second flared region **15**. In some embodiments, the outer radial extent and/or the radially outermost extent of the plurality of anti-migration supports **20** and/or the anti-migration loops in the radially expanded configuration and/or at the first angle $\theta_{sub.1}$ may be within about $\pm 10\%$, about $\pm 5\%$, about $\pm 3\%$, about $\pm 1\%$, or another suitable range of the outer diameter of the first flared region **13** and/or the second flared region **15**.

[0068] Those skilled in the art will recognize that the present invention may be manifested in a variety of forms other than the specific embodiments described and contemplated herein. Accordingly, departure in form and detail may be made without departing from the scope and spirit of the present invention as described in the appended claims.

[0069] The materials that can be used for the various components of the esophageal stent(s) and the various elements thereof disclosed herein may include those commonly associated with medical devices and/or stents. For simplicity purposes, the following discussion refers to the apparatus. However, this is not intended to limit the devices and methods described herein, as the discussion may be applied to other elements, members, components, or devices disclosed herein, such as, but not limited to, the esophageal stent, the expandable framework, the filaments, the anti-migration supports or loops, the covering, the outer sleeve and/or elements or components thereof.

[0070] In some embodiments, the apparatus, and/or components thereof, may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material.

[0071] Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), MARLEX® high-density polyethylene, MARLEX® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro(propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, polyurethane silicone copolymers (for example, ElastEon® from Aortech Biomaterials or ChronoSil® from AdvanSource Biomaterials), biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP). For example, the mixture can contain up to about 6 percent LCP.

[0072] Some examples of suitable metals and metal alloys include stainless steel, such as 304V, 304L, and 316LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or

super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; platinum; palladium; gold; combinations thereof; or any other suitable material.

[0073] In at least some embodiments, portions or all of the apparatus, and/or components thereof, may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids the user of the apparatus in determining its location. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the apparatus to achieve the same result.

[0074] In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the apparatus and/or other elements disclosed herein. For example, the apparatus, and/or components or portions thereof, may be made of a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The apparatus, or portions thereof, may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nitinol, and the like, and others.

[0075] In some embodiments, the apparatus and/or other elements disclosed herein may include and/or be treated with a suitable therapeutic agent. Some examples of suitable therapeutic agents may include anti-thrombogenic agents (such as heparin, heparin derivatives, urokinase, and PPACK (dextrophenylalanine proline arginine chloromethylketone)); anti-proliferative agents (such as enoxaparin, angiopeptin, monoclonal antibodies capable of blocking smooth muscle cell proliferation, hirudin, and acetylsalicylic acid); anti-inflammatory agents (such as dexamethasone, prednisolone, corticosterone, budesonide, estrogen, sulfasalazine, and mesalamine); antineoplastic/antiproliferative/anti-mitotic agents (such as paclitaxel, 5-fluorouracil, cisplatin, vinblastine, vincristine, epothilones, endostatin, angiostatin and thymidine kinase inhibitors); anesthetic agents (such as lidocaine, bupivacaine, and ropivacaine); anti-coagulants (such as D-Phe-Pro-Arg chloromethyl ketone, an RGD peptide-containing compound, heparin, anti-thrombin compounds, platelet receptor antagonists, anti-thrombin antibodies, anti-platelet receptor antibodies, aspirin, prostaglandin inhibitors, platelet inhibitors, and tick antiplatelet peptides); vascular cell growth promoters (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional activators, and translational promoters); vascular cell growth inhibitors (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional repressors, translational repressors, replication inhibitors, inhibitory antibodies, antibodies directed against growth factors, bifunctional molecules consisting of a growth factor and a cytotoxin, bifunctional molecules consisting of an antibody and a cytotoxin); cholesterol-lowering agents; vasodilating agents; and agents which interfere with endogenous vasoactive mechanisms.

[0076] It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the invention. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A stent, comprising: an expandable framework including a body region formed of one or more interwoven filaments, the expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, wherein the expandable framework is configured to expand from a radially collapsed configuration to a radially expanded configuration; wherein the expandable framework includes a plurality of anti-migration supports extending radially outward from the body region; a polymeric covering encapsulating the expandable framework and the plurality of anti-migration supports; and a polymeric outer sleeve spaced apart radially outward from the body region of the expandable framework by the plurality of anti-migration supports.
2. The stent of claim 1, wherein the polymeric outer sleeve is fixedly attached to the polymeric covering proximate the first end and the second end of the expandable framework.
3. The stent of claim 1, wherein the polymeric outer sleeve is fixedly attached to the polymeric covering at a radially outermost extent of the plurality of anti-migration supports.
4. The stent of claim 1, wherein the plurality of anti-migration supports extends radially outward from the body region at a first oblique angle to the central longitudinal axis.
5. The stent of claim 1, wherein the plurality of anti-migration supports is configured to elastically deflect radially inward to a second oblique angle to the central longitudinal axis less than the first oblique angle when subjected to an external force applied in a radially inward direction.
6. The stent of claim 5, wherein the external force is greater than about 0.5 Newtons and less than about 5.0 Newtons.
7. The stent of claim 5, wherein the body region of the expandable framework is configured to maintain a substantially constant inner diameter when subjected to the external force.
8. The stent of claim 1, wherein the plurality of anti-migration supports is substantially rigid from a base of the plurality of anti-migration supports to a radially outermost extent of the plurality of anti-migration supports.
9. A stent, comprising: an expandable framework including a body region formed of one or more interwoven filaments, the expandable framework having a first end, a second end, and a central longitudinal axis extending from the first end to the second end, wherein the expandable framework is configured to expand from a radially collapsed configuration to a radially expanded configuration; wherein the expandable framework includes a plurality of anti-migration supports extending radially outward from the body region; a polymeric covering encapsulating the expandable framework and the plurality of anti-migration supports; and a polymeric outer sleeve spaced apart radially outward from the body region of the expandable framework by the plurality of anti-migration supports; wherein the polymeric covering and the polymeric outer sleeve define a plurality of spaces therebetween.
10. The stent of claim 9, wherein the plurality of spaces is a plurality of discrete chambers.
11. The stent of claim 9, wherein the plurality of spaces is interconnected between the polymeric covering and the polymeric outer sleeve.
12. The stent of claim 9, wherein a base of each anti-migration support of the plurality of anti-migration supports forms a hinge portion with respect to the body region.
13. The stent of claim 9, wherein the plurality of anti-migration supports is self-biased toward extending radially outward from the body region at a first oblique angle to the central longitudinal

axis when not subjected to an external force applied in a radially inward direction.

14. The stent of claim 13, wherein the plurality of anti-migration supports is configured to elastically deflect radially inward to a second oblique angle to the central longitudinal axis less than the first oblique angle when subjected to the external force applied in the radially inward direction.

15. A stent, comprising: an expandable framework including a body region formed of one or more interwoven filaments, the expandable framework having a first flared end region, a second flared end region, and a central longitudinal axis extending from the first flared end region to the second flared end region, wherein the expandable framework is configured to expand from a radially collapsed configuration to a radially expanded configuration; wherein the expandable framework includes a plurality of anti-migration supports extending radially outward from the body region; a polymeric covering comprising: a first portion encapsulating the first flared end region, the second flared end region, and the body region; and a second portion encapsulating the plurality of anti-migration supports; wherein the polymeric covering spans interstices between the one or more interwoven filaments; and a polymeric outer sleeve spaced apart radially outward from the body region of the expandable framework by the plurality of anti-migration supports; wherein the polymeric outer sleeve is fixedly attached to the first portion of the polymeric covering at the first flared end region and the second flared end region of the expandable framework.

16. The stent of claim 15, wherein the first portion of the polymeric covering has a first Shore hardness and the second portion has a second Shore hardness different than the first portion.

17. The stent of claim 16, wherein the second Shore hardness is higher than the first Shore hardness.

18. The stent of claim 15, wherein each anti-migration support of the plurality of anti-migration supports comprises a base hingedly coupled to the body region and a tip disposed opposite the base.

19. The stent of claim 18, wherein the polymeric outer sleeve is fixedly attached to the second portion of the polymeric covering at the tip of each anti-migration support of the plurality of anti-migration supports.

20. The stent of claim 15, wherein each anti-migration support of the plurality of anti-migration supports is oriented at an oblique angle to the central longitudinal axis extending toward the second end.
